Appendix F

The Photon Model of Polarizing Information

The electrodynamic framework developed in Chapters 2–21 treats the Interference Engine as a continuous field: a superposition of cultural magnetic fields B_k with time-varying demographic weights $\rho_k(t)$. This appendix extends the field model into the quantized regime by treating discrete information packets as photon-like carriers of polarization.

The extension is speculative. It is presented as a formal analogy with predictive value, not as a claim that social media posts are literal electromagnetic quanta.

F.1 From Continuous Field to Discrete Quanta

In the continuous model, the Engine injects power into the k-th identity axis at rate:

$$P_k = \frac{1}{2\mu_0}|B_k|^2\, V \tag{F.1}$$

where V is the effective volume of the population exposed to the field. In the analog regime, V was bounded by physical reach; in the digital regime, V approaches the entire network.

The continuous model predicts total power; the photon model quantizes the distribution of that power across individual exposures. Define a **polarizing photon** as a discrete information packet (tweet, meme, short-form video, algorithmic recommendation) that carries a specific polarization state along one or more identity axes.

F.2 The Photon Ansatz

A polarizing photon has the following properties:

1. **Energy**: $E_\gamma = h\nu$, where ν is the refresh/repetition frequency of the axis and h is the attention-economy constant. High-frequency axes (e.g., transphobia cycles in 2023) have higher photon energy than low-frequency axes (e.g., racial housing covenants, which operated on decadal cycles).

2. **Polarization**: Each photon carries a spin state $\sigma_k \in \{-1, 0, +1\}$ along axis k. The spin determines whether the packet reinforces Buffer Class alignment $(+1)$, Out-group extraction (-1), or neutral transmission (0).

3. **Propagation velocity**: $c_{\text{effective}}$, the network latency. In practice, this is milliseconds for domestic propagation and seconds for global propagation—effectively instantaneous at human perception scales.

4. **No rest mass**: The photon exists only in motion. Once shared, it propagates until absorbed (viewed, processed, emotionally reacted to) or decohered (drowned by competing signals).

F.3 Spectral Density and the Small-Demographic Anomaly

The photon model explains the small-demographic targeting puzzle (Section VI of the electrodynamic synthesis) with precision. The total power P_k injected into axis k can be written as:

$$P_k = N_\gamma \cdot E_\gamma = N_\gamma \cdot h\nu_k \tag{F.2}$$

where N_γ is the photon number (packet count) and ν_k is the axis frequency.

For a major axis like race (B_1), the population base is large. To avoid saturating the Buffer Class (burnout / legitimacy collapse), the Engine must distribute the same power across many low-energy photons: high N_γ, low ν_k. The field is spatially diffuse.

For a minor axis like transphobia (B_3), the population base is small ($\sim$0.6%). The Engine can concentrate the same power into fewer photons of higher individual energy: low N_γ, high ν_k. The Buffer Class has been pre-conditioned by patriarchal B_{gender} to exhibit high magnetic susceptibility χ_m on this axis, so the high-energy photons achieve the same deflection at lower total packet count.

The result is a **spectral optimization**: the Engine selects the axis frequency ν_k and photon number N_γ that maximize deflection per unit cost:

$$\max_{k,N_\gamma} \chi_{m,k} \cdot N_\gamma \cdot h\nu_k \quad \text{subject to} \quad C_{\text{engine}}^{\text{digital}}(N_\gamma, \nu_k) \leq C_{\text{budget}} \tag{F.3}$$

This is why a demographic of 0.6% can command a field energy comparable to a demographic of 13%. The energy is distributed according to the spectral efficiency of deflection, decoupling field share from population proportion.

F.4 Implications for Counter-Engineering

If the interference is carried by photon-like packets, then resistance can be modeled as **polarization filtering**. The Counter-AI (Chapter 23) must function as a polarizing filter that selectively attenuates high-ν_k photons while amplifying class-band photons (f_{class}).

The analogy suggests specific engineering requirements:

- **Frequency-selective attenuation**: Detect and suppress packets whose ν_k exceeds the natural frequency of organic political discourse. Machine-generated high-frequency polarization (botnets, synthetic media) should be filtered before human exposure.

- **Polarization rotation**: Convert orthogonal deflections back into class-aligned momentum. When a packet with $\sigma_k = +1$ (racial resentment) is detected, the Counter-AI publishes the corresponding class-axis packet showing how E benefits from the same resentment.

- **Coherent amplification**: Class-band photons must be amplified constructively. The Counter-AI must increase N_γ at f_{class} until the class-band power exceeds the identity-band power—reversing the spectral redistribution documented in Chapter 22.